Cognition and Decision Making

June 10, 2025

Background material to the choice prediction competition (CPC) that will serve as the protective project in the course.

This current text is based on a grant proposal submitted by Ido Erev to NSF in January, 2025. The title of the proposal is "**Decisions Based on Potentially Misleading Descriptions".**

The class's CPC will be like the study proposed in Section 3.4 of the proposal. The baseline model will be like the model described in Section 2.3

Your main task:

On July 10 2025, at 8 am you will receive an excel file that includes at least 100 conditions, with 8 missing values in each row. Your task is to predict the missing values, and send us an excel with the predicted value by 14:00 on the same day. Here is an example of two possible rows.

a1	pa1	a2	corrAB	b1	pb1	b2	A_label	B_label	Forgone	n	1shot	Chat GPTo	BEAST-GB	arate1	arate2	arate3	arate4	afoka	afrega	afregb	afokb
0	1	0	0	10	0.1	-1	Maintains the status quo	Can lead to a large gain	0	23	0.23	0.215	.21								
0	1	0	0	8	0.1	-1	Maintains the status quo	Can lead to a large gain	0	25	0.23	0.206	.31								

To train your model to predict these values you can use the training data set (in the class's moodle) that includes more than 200 conditions from the same space of condition. The model should read the predictors (a1 to BEAST-GB) and predict the 8 missing values (artae1-afokb). The names of the variables are in the "competition variables names" file in the class's moodle

1. Introduction

Mainstream decision research (e.g., Kahneman & Tversky, 1979) primarily focuses of the way people decide when they can rely on an accurate numerical description of the incentive structure. In contrast, many real-world decisions are made based on verbal descriptions that can include both guidance and misguidance. Guidance is defined here as a description that directs or nudges decision-makers toward options that maximize their expected value (EV). Examples of guidance include road signs like "dangerous curve ahead" and ingredient lists on cereal boxes. Misguidance, on the other hand, refers to descriptions that lead decision-makers away from optimal choices (I use the terms misguidance and misleading information interchangeably). Advertisements promoting state lotteries or unhealthy foods are examples of misguidance. The current proposal seeks to extend foundational decision research by clarifying how people make decisions based on verbal descriptions of this kind.

The starting point of the present analysis is the assumption that people distinguish between guidance and misguidance by drawing on two types of past experiences: previous encounters with similar descriptions, and direct experiences related to the current choice task. For example, when I recently attempted to enter an unfamiliar supermarket at 7:30 a.m., I had to choose between a door labeled "Entry" and another labeled "Exit, no entry." Drawing on past experiences, I chose the Entry door. To my surprise, it was locked, and I was directed to use the other door. On my next visit, at 7:50 a.m., the following day, I tried the Exit door first.

The implications of this assumption are developed here by building on two sets of observations. The first involves studies showing that, in certain contexts, large language models (LLM) can offer reasonable predictions of initial reactions to verbal descriptions (see Binz et al., 2024; Marjieh et al., 2024). The second set of supporting observations stems from basic research indicating that the influence of experience on choice behavior follows general cognitive processes. This research is reviewed in Section 2 (Background and Pilot Studies).

The proposed research addresses five related goals. The first goal is to generalize the study of one-shot decisions, expanding from accurate numerical descriptions to include a wider variety of descriptions. Pilot results suggest a significant difference between the models that best predict decisions based on numerical and verbal descriptions. While the best predictors for decisions based on verbal descriptions are pure large language models, which do not abstract the underlying cognitive processes, the best predictors for the initial reaction to numerical descriptions involve abstractions of those processes. I plan to evaluate the robustness and implications of this finding by organizing an open-choice prediction competition, challenging other researchers to provide better models.

The second goal is to clarify how feedback (new experiences) modifies reactions to descriptions. This analysis begins by testing the hypothesis that the basic properties of human responses to experience, as documented in previous research on decisions based on numerical descriptions and given initial ignorance, also apply when decision-makers rely on verbal descriptions.

The third goal is to evaluate non-intuitive predictions derived from a simple model assuming that the processes driving the effects of experience are independent of the type of description. Examples of such predictions include similar sensitivity to guidance and misguidance even after 100 trials with immediate feedback, and the identifying of conditions under which experience increases the tendency to trust misleading fake news, and causes well-intended warnings to backfire.

The fourth goal is to compare alternative theoretical approaches for deriving useful quantitative predictions of the combined impact of distinct descriptions and experience. I hypothesize that behavior in these settings can be predicted by using large language models to estimate the initial reaction to new verbal descriptions, and then approximating how experience modifies this initial reaction by employing a cognitive model that captures three general effects of experience. This hypothesis will be tested by organizing a second open-choice prediction competition.

The fifth goal is to clarify the conditions under which abstracting cognitive processes improves the predictions of large language models and other machine learning tools.

2. Background and Pilot Studies

As noted earlier, mainstream decision research focuses on one-shot decisions based on precise descriptions of the incentive structure. For instance, in one of the tasks studied by Kahneman and Tversky (1979), participants were asked to choose between receiving "5 for sure" or taking "a 1 in 1000 chance to win 5000; 0 otherwise." This approach has enabled the identification of robust deviations from the predictions of the rational decision model and fostered the development of elegant descriptive models that capture these deviations. The basic rational model, the expected value rule, prescribes weighting possible outcomes by their probabilities. The leading descriptive models generalize this model and assume that people weight subjective functions of outcomes by subjective functions of probabilities (Wakker, 2010).

Extending this influential research to decisions based on potentially misleading verbal descriptions presents significant challenges. When descriptions are ambiguous or misleading, almost any behavior can be rationalized under some interpretation of the available information. Consequently, the rational model fails to provide clear predictions, and insights into deviations from rational choice offer limited guidance in such cases. The current proposal seeks to address this challenge by building on two lines of research: (1) studies that employ machine learning tools to predict human behavior and (2) research on decisions made from experience. The following sections review these two research streams and present pilot experiments (conducted with Ori Plonsky, Yefim Roth, and Eyal Marantz) designed to clarify the relevance of these research streams to the current proposal.

2.1 Predicting decisions from descriptions using machine learning and cognitive models

It is useful to distinguish between four approaches to developing descriptive models of choice behavior. The first approach focuses on modeling the underlying cognitive processes. For example, the BEAST model (Best Estimate And Sampling Tools), which won the 2015 choice prediction competition (CPC15, Erev et al., 2017), assumes that individuals choose the most attractive option, where attractiveness is the sum of the option's EV and the payoff it obtained in a small sample of past experiences. A second approach refines the predictions of cognitive models using machine learning. For instance, the BEAST-GB model, which won the 2018 choice prediction competition (CPC18, Plonsky et al., 2024), uses gradient boosting (Friedman, 2001) to enhance the predictions of BEAST. A third approach uses machine learning to quantify inputs to a model of cognitive processes. For example, Gandhi et al. (2022) demonstrate how large language models can be used to quantify human perceptions of the healthiness of different foods. The fourth approach uses large language models without abstracting the underlying cognitive processes. Binz et al. (2024) show that, in certain settings, the large language model they developed (Centaur) outperforms leading cognitive models. Pilot Study 1 sheds light on the value of these approaches in the context of one-shot decisions.

2.1.1 Pilot Study 1

This study focused on 1,000 randomly generated pairs of binary prospects. A verbal description of each pair was generated using ChatGPT. Three of these pairs, along with their verbal descriptions, are presented in Table 1. (Note a in Table 1 explains the random generation algorithm, and Note b describes the prompt to ChatGPT.)

Table 1: Examples of three of the choice pairs examined in Pilot Study 1, their description, and the main results.

Examples of choice pairs[a]	The verbal descriptions generated by ChatGPT[b]	Main results Choice rate of Option A	
		Observed	RoBERTa
A. 3 for sure B. 96% to win 9; 4% to lose 90	A. A definite small amount B. A strong chance of gaining some, a small chance of losing a lot	.47	.27
A. 1 for sure B. 4% to win 58; 96% to lose 5	A. This is an absolute chance of a gain B. The chances of losing far outweight the chance of gaining	.76	.93
A. 88% to win 1, 18% to loss 2 B. 61% to win 40; 39% to lose 73	A. A small chance to lose a couple B. Middle ground chance at a middle number	.21	.38

[a] The random selection algorithm was similar to the one used in Erev et al. (2017), with the additional constraint that one option was a safe prospect with one or two outcomes, while the other was a risky option with two outcomes. The payoff range was from -99 to +99. In 14% of the tasks, one option was the dominant choice.

[b] The prompts for ChatGPT were variants of the following basic prompt: "Each gamble in the dataset is associated with a set of descriptive labels. I need help creating descriptions for these gambles based on their labels. Each label set should be transformed into a coherent and engaging sentence that conveys the essence of the gamble, with the context of all labels integrated either implicitly or explicitly. The descriptions should:

- Be clear and accessible to a general audience, without requiring expert knowledge.
- Avoid technical jargon, aiming for an engaging and informative style.
- Vary in language and structure to avoid repetition.
- Provide a holistic understanding of each gamble, reflecting the broader context provided by all the labels.
- Avoid numerical specifics, ensuring the description remains grounded in realism and context."

The experiment examined one-shot decisions based on verbal descriptions. Figure 1 presents the general instructions and a typical trial.

Figure 1: The one-shot decisions from description paradigm.

Instructions: In each trial of the current experiment you are asked to select one of two options that can lead to a gain or a loss of points. Your initial balance is 1000 points. Your final point balance will determine the probability of winning a bonus of $0.5 (in addition to the base participation payment of $0.4).
Example of a choice task. Choose between: A. a definite small amount B. a strong chance of gaining some, a small chance of losing a lot

Participants and procedure.
The participants were 1107 Cloud Research workers. Each participant faced 20 (randomly selected) of the 1000 pairs. All the participants received a show-up fee of $0.4 for an experiment that lasted less than 5 minutes, and as explained in Figure 1 could win an additional bonus (of $0.5) depending on their performance (this payment rule implies that to maximize the probability of getting a bonus, participants should aim to maximize the average number of points across all trials). Each task was faced by 20 to 24 participants.

Analysis and results
The analysis focused on comparing the four modeling approaches described above. To estimate the models' prediction value, they were first estimated based on a training set comprising 90% of the tasks, and then their accuracy (Means Squared Error, MSE) was computed on a test set containing the remaining 10%. The best model that we could find is a large language model fine-tuned to the training set: The model RoBERTa (Liu et al., 2019). (Tuning of other LLMs we tried, including GPT4o, was slightly less accurate; other approaches were significantly less accurate). RoBERTa's MSE score is 0.1830. The minimal possible score (the variance) is 0.1735. Thus, the bias of the model is 0.1830-0.1735 = 0.0095.

To clarify the implications of this finding, we performed two additional analyses. First, we derived the MSE and the bias of the model BEAST-GB when predicting one-shot decisions based on numerical description in the large "Choices13k" data set (Bourgin et al., 2019; Peterson et al., 2021); the MSE is 0.214 and the bias is .0103. This analysis suggests that the predictive value of RoBERTa in capturing decisions from verbal descriptions, is similar to the predictive value of BEAST-GB in capturing decisions based on numerical descriptions. In a second analysis we tried to find a large language model that outperforms BEAST-GB in predicting decisions based on numerical descriptions. This effort was not successful. The MSE of RoBERTa in predicting the Choices13k data is 0.2247 and the bias is 0.0230 (while the bias on BEAST-GB is only 0.0103); we could not find a large language model that provides better predictions.

2.2 Three apparent deviations from adaptive learning
Analyses of the impact of immediate feedback on choice behavior highlight a general tendency to repeat choices that were found to provide the best average payoff (in accordance with the law of effect, Thorndike, 1898) and several apparent deviations from this tendency. These deviations include: underweighting of rare outcomes (Barron & Erev, 2003), the correlation effect (Diederich & Busemeyer, 1999), and a surprise-trigger-change effect (Nevo & Erev, 2012). Importantly, these effects were observed under two very different prior information conditions: in studies of decisions based on precise

and accurate numerical description (when the experience does not add objectively useful information), and also in studies of decisions under initial ignorance (when the experience is the sole source of information). In order to clarify the presentation of these effects, and their relationship to the proposed research, I chose to run a pilot study that replicates them using the experimental paradigm I plan to use in the proposed research.

2.2.1 Pilot Study 2.

Pilot study 2 used the experimental paradigm described in Figure 2, and examined repeated decisions. It focused on the four pairs choice tasks described in Figure 3. Each pair was studied in the absence of description (Condition Blank) and based on precise numerical description (Condition Numerical). Figure 2 presents the full instructions, and the main screens in Condition Numerical.

Figure 2: The repeated decisions paradigm.

Instructions: This is a decision-making experiment consisting of 100 rounds. It is expected to take about 10 minutes. In each round, you will have to choose between keys that will remain the same throughout the game.
You will be able to base your decision on two sources of information: a description of the possible options, and feedback that will be presented after each choice. The feedback will show the outcome (in points) from the alternative you chose, along with the outcomes you would have gotten had you chosen the other alternatives. Your base payment for participating in this experiment is $1.5. In addition, you can win a bonus of $0.5. Your initial endowment is 1000 points. Earning points increases the probability of a bonus, and losing points reduces this probability.

Example of a pre-choice screen:

Total number of points: 1000

Please choose one of the keys:

0 for sure	10% chance to lose 10, and 90% chance to win 1

Example of a feedback screen (after a choice of the Right key):

Total number of points: 1001

0 for sure	10% chance to lose 10, and 90% chance to win 1
0	**+1**

You selected Right, and earned 1 point
The payoff from selecting the other key is presented in Gray

Note: The upper panel presents the initial instructions. The middle and lower panels present typical pre-choice and feedback screens in condition that examine decisions based on numerical description with full feedback.

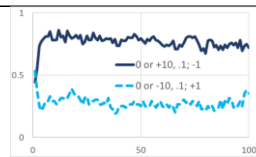
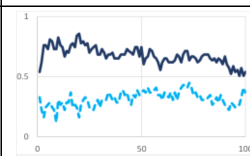
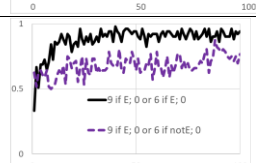
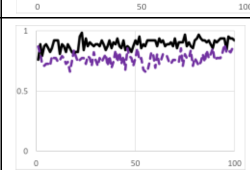
Participants and procedure.

At least 50 Cloud Research workers participated in each of the 8 conditions. Each participant faced one choice task for 100 trials with immediate feedback after each choice. The experiment lasted about 10 minutes. As explained in Figure 2, all the participants received a show-up fee of $1.5, and could win an additional bonus.

Results
The study of Pairs 1 and 2 in Condition Blank reveals significant deviations from maximization: The safer option (0 for sure) was selected in most trials in Pair 1 when the EV of risky option was positive (EV = +0.1), but in less than 50% of the trials in Pair 2 when the EV of risky option was negative (EV = -0.1). Both deviations can be captured by assuming underweighting of the rare (0.1) outcomes (Barron & Erev, 2003). Condition Numerical shows that the presentation of description reduces, but does not eliminate the tendency to underweight the rare outcomes.

The study of Pairs 3 and 4 in Condition Blank reveals the correlation effect (Diedrich & Busemeyer, 1999): Quick learning to maximize when the payoffs are positively correlated (Pair 3), and slow learning in the face of negative correlation (Pair 4). Condition Numerical highlights the robustness of this pattern, and also shows that it leads to an initial learning away from the optimal choice.

Figure 3: Summary of Pilot Study 2. Each of the four choice tasks was studied under two description conditions.

	Payoff distributions		Observed choice rate of Option A as a function of time (trial 1 to 100)			"Repeating A" rate after a trial in which		"Switching to A" rate after a trial in which	
Pair	Option A	Option B	Blank	Numerical	Desc.	A paid more (afoka)	B paid more (afrega)	A paid more (afregb)	B paid more (afokb)
1	0 for sure	+10, .1; -1 otherwise			Blnk	0.91	0.88	***0.15***	***0.58***
					Num	0.90	0.78	***0.27***	***0.53***
2	0 for sure	-10, .1; +1 otherwise			Blnk	***0.39***	***0.54***	0.31	0.12
					Num	***0.61***	***0.81***	0.12	0.10
3	9 if E; 0 if notE	6 if E; 0 if notE			Blnk	0.92		0.67	
					Num	0.93		0.58	
4	9 if E; 0 if notE	6 if notE 0 if E;			Blnk	0.86	0.83	0.62	0.41
					Num	0.82	0.72	0.56	0.42

Note: The notation x, p; y implies "earn x with probability p; earn y otherwise." The notation x if E; y if notE" implies "earn x if Event E occurs; earn y otherwise." The probability of event E in the current conditions was 0.5. Desc, Blnk, and Num, imply Description, Blank, and Numerical, respectively.

Analysis of the sequential dependencies (right side of Figure 3) reveals positive recency in most cases (higher choice rate of Option A after trials in which A paid more), but four clear reversals of this pattern (the pairs of ***bold italic*** cells). Nevo and Erev (2012) show that this pattern can be captured by assuming a surprise-trigger-change effect: A tendency to repeat the last choice that decreases when the feedback is surprising.

2.3 The inertive Sampler (iSampler) model

The phenomena illustrated in Figure 3 can be captured through a generalization of the model greedy sampler (Erev et al., 2023). The greedy sampler assumes that each choice is influenced by either old experiences (obtained before the current experiment) or new experiences (from previous trials in the current experiment). The probability of relying on old experiences is 1 in the first trial, and it gradually decreases toward an asymptote determined by δ_i, a free parameter that captures the tendency of agent *i* to rely on old experiences. Specifically, the probability of relying on old experiences after observing an average of *t* outcomes from each option is expressed as: P(Old) = $\delta_i^{(t-1)/t}$.

The greedy sampler model was originally developed to account for the effects of experience in the absence of descriptive information (Condition Blank). Under these conditions, reliance on old experiences is approximated by assuming indifference (random choice) between the two options. In other trials, the model assumes an effort to rely on the most similar κ_i (a free parameter) past trials. Plonsky et al. (2015) demonstrate that in dynamic settings, this effort can be highly adaptive. However, in the current static setting, where all past trials are equally similar, this effort is counterproductive, and its impact can be approximated by assuming reliance on a random sample of κ_i past trials.

The iSampler model extends the greedy sampler model in two significant ways. First, it approximates the impact of reliance on old experiences by building on the models described in Pilot Study 1. Specifically, RoBERTa is used for verbal descriptions, and BEAST-GB is applied for numerical data. This extension implies that iSampler quantifies the inputs to the cognitive model using machine learning. Second, to account for the surprise-trigger-change effect, iSampler introduces inertia, defined as a tendency to repeat the last choice when outcomes are not surprising (as defined below). The probability that agent *i* will make a new choice at trial *t* + 1 (rather than exhibiting inertia) is η_i (a free parameter) if the last payoff is not surprising and 1 otherwise.

The probability of surprise is calculated as follows: Let $GM_{j,t}$ denote the grand mean (average payoff) from *j* over the first *t* trials, $V_{j,t}$ the payoff from *j* in trial *t*, and Max_j and Min_j the highest and lowest experienced payoffs from prospect *j*. The probability that the payoff from *j* in trial *t* triggers surprise is:

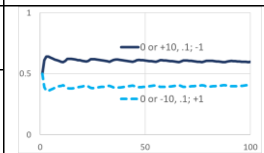
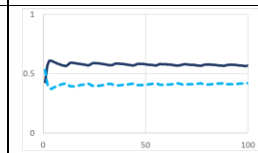
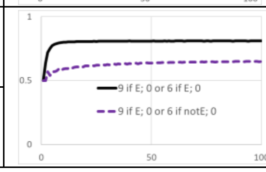
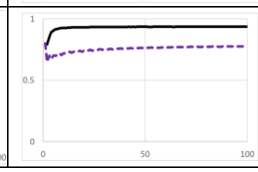
$$P(j\ triggers\ surprise\ at\ t) = \begin{cases} 0 & if\ \mathrm{Max_j} = \mathrm{Min_j} \\ \left| GM_{\mathrm{j,t}} - \mathrm{V_{j,t}} \right| \big/ [\mathrm{Max_j} - \mathrm{Min_j}] & otherwise \end{cases}$$

When the agent does not make a new choice, they repeat their last choice.

iSampler was estimated based on the 90 experimental conditions used by Erev et al. (2023) to estimate the greedy sampler. Consistent with Erev et al., the estimation was independently conducted for each of the 3,701 participants in these conditions. The output of this procedure is an estimated distribution of the model's three parameters across the population. The model's predictions for new conditions were derived using this estimated distribution without re-estimating the parameters. A comparison of the observed (Figure 3) and reproduced results (presented in Figure 4) shows that iSampler effectively captures the three robust effects of experience described above.

Additional analysis reveals that iSampler provides equally accurate predictions for decisions based on numerical descriptions (the 150 conditions examined in Erev et al., 2017) and for pure decisions from experience with partial feedback (the 120 repeated decision conditions examined by Erev et al., 2010, when feedback is limited to the outcome of the selected option).

Figure 4: The predictions of iSample for Pilot Study 2.

	Choice tasks		Predicted choice rate of Option A as a function of time (trial 1 to 100)			"Repeating A" rate after a trial in which		"Switching to A" rate after a trial in which	
Pair	Option A	Option B	Blank	Numerical	Desc.	A paid more (afoka)	B paid more (afrega)	A paid more (afregb)	B paid more (afokb)
1	0 for sure	10, .1; -1 otherwise			Blnk	0.85	0.64	***0.26***	***0.38***
					Num	0.84	0.63	***0.25***	***0.37***
2	0 for sure	-10, .1; 1 otherwise			Blnk	***0.62***	***0.75***	0.36	0.15
					Num	***0.65***	***0.76***	0.38	0.16
3	9 if E; 0 if notE	6 if E; 0 if notE			Blnk	0.86		0.60	
					Num	0.95		0.75	
4	9 if E; 0 if notE	6 if notE 0 if E;			Blnk	0.74	0.71	0.50	0.44
					Num	0.85	0.82	0.56	0.48

Note: Same notations as in Figure 3.

Evaluation of the distribution of the estimated parameters reveals substantial individual differences. The parameters of approximately 20% of participants suggest a tendency to trust new experiences (δ_i close to 0), while the parameters of about 10% of the participants suggest a tendency to rely only on the description and ignore new experiences (δ_i close to 1). Similarly, the estimation of κ_i (the sample size) reveal high variability: The choices of most participants are best fitted with κ_i below 7, but about 15% of the participants behave as if they rely on all previous trials.

3. The proposed studies

3.1. Study 1: Comparison of alternative models of one-shot decisions

Pilot Study 1 highlighted a puzzling difference between one-shot decisions based on verbal and numerical descriptions. Specifically, the best models found in this study capture decisions from verbal descriptions without explicit abstraction of the underlying cognitive process, but use an abstraction of these processes to capture the decisions from numerical descriptions. One explanation for this finding suggests a qualitative difference between the processes underlying decisions based on verbal versus numerical descriptions. Specifically, decisions derived from numerical descriptions may be more reliant on straightforward, serial cognitive computations that are easier to model. An alternative explanation posits that this difference arises because the large language models considered in Pilot Study 1 were trained predominantly on natural text, which contains far more verbal than numerical descriptions. Thus, training large language models on a more extensive dataset of experimental studies involving numerical descriptions might eliminate the observed predictive advantage of explicitly abstracted cognitive processes. A third explanation suggests that the difficulty in identifying a cognitive model that facilitates the prediction of the reaction to verbal descriptions may stem from the limited range of cognitive models explored.

Study 1 aims to evaluate these hypotheses by organizing an open choice prediction competition focusing on one-shot decisions based on three types of descriptions: Numerical, Verbal descriptions generated by ChatGPT (as in Pilot Study 1), and Verbal descriptions generated by human participants (that will be collected in preliminary "description" experiments). As in Erev et al. (2010, 2017), the competition will follow the five-step format outlined in Table 2 (Plonsky & Erev, 2023).

Table 2: The five-step format of the proposed choice prediction competitions

Steps
1. Selecting the **focus** of the competition: the experimental paradigm, a well-defined space of choice tasks (the "competition space"), the prediction goal (the statistics to be predicted), and the accuracy criterion (the current analysis uses MSE).
2. Running a large **training experiment** on randomly selected choice tasks drawn from the competition space.
3. Proposing a **baseline model**.
4. Posting **a call** that challenges other researchers to submit models to a competition that focuses on predicting behavior in a target study that examines behavior in a new set of tasks drawn from the competition space. The call includes a submission deadline, and links to the training dataset and the baseline model.
5. After the submission deadline: Running the **target experiment** and computing the MSE of the submitted models.

The **focus** of Study 1 will be the decisions-from-description paradigm (Figure 1), the space of choice tasks examined in Pilot Study 1, and the choice rate as the prediction goal.

Training, target, and description experiments (Participants, Design, Task, and Procedure): The training and target experiments will follow the methodology of Pilot Study 1 but will encompass a broader set of choice tasks. The training and target experiments will comprise 4,000 pairs of prospects each, with each pair tested under three description conditions. A combination of a pair and description will be referred to as a "choice task." Each choice task will be presented to 20 participants, and each participant will face 20 tasks. Thus, the total number of participants in the main part of the training and target experiments will be (2)(4,000)(3)(20)/20 = 24,000.

Each participant in the preliminary description experiments will describe 5 pairs, and the instructions will be similar to the prompt used in Pilot Study 1 (Note b in Table 1). The total number of participants for these experiments will be (2)(4,000)/5 = 1600. Like the participants in the other experiments, these participants will be recruited from Cloud Research, and will be compensated $1.5 for a task lasting approximately 10 minutes. If the Cloud Research participant pool proves insufficient, I will consider additional participant recruitment platforms.

Baseline model and predictions: The baseline model will utilize fine-tuned RoBERTa for verbal descriptions and BEAST-GB for numerical descriptions. As noted earlier, the study compares alternative hypotheses regarding the nature of the winning submissions.

Competition Call: The competition call will be posted six months before the submission deadline on leading email lists in decision research, behavioral economics, machine learning, and reinforcement learning. The call will include a link to the competition website, which will host a paper summarizing the competition focus, methodology, the results of the training experiment, and baseline model. Additionally, the site will provide access to raw data and participation rules (including the deadline, submission method, and submission format requirements). The winner will receive co-authorship of the paper summarizing the competition.

3.2. Study 2: The impact of verbal descriptions on the reaction to feedback

Study 2 tests the hypothesis that the three effects of experience demonstrated in Pilot Study 2 also emerge in decisions based on verbal descriptions. In order to achieve this goal, it examines how the modification of the descriptions changes the impact of experience on decisions between the choice pairs used in Pilot Study 2.

Participants, Design, Task, and Procedure: In the first part of Study 2, each of the four choice pairs from Figure 3 (Pilot Study 2) will be described by 10 human participants (using the method of the description experiments of Study 1). This process will yield 40 sets of choice tasks and descriptions.

In the main part of the study, 20 participants will face each of the 40 sets using the methodology from Pilot Study 2. Thus, the total number of participants in the main part will be 800, with each participant encountering one choice task over 100 trials.

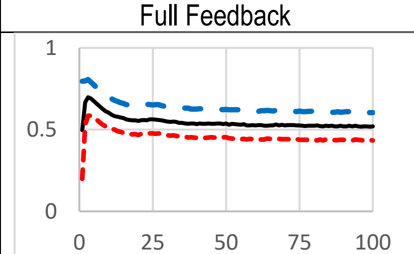
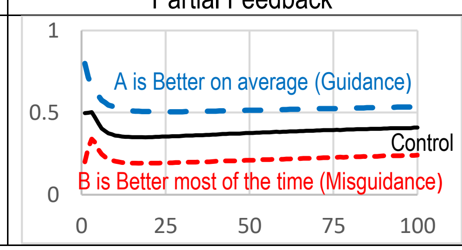
Predictions: The robustness of the three effects of experience documented in Pilot Study 2 suggests similar tendencies in response to verbal descriptions. Thus, I predict: (1) Underweighting of rare outcomes: The maximization rate for the distinct descriptions of Pairs 1 and 2 will be below 50%. (2) Correlation effect: The maximization rate for the distinct descriptions of Pair 4 will be higher than that for Pair 3. And (3) Surprise-trigger-change pattern: Negative recency following risky choices in Pair 1 and safe choices in Pair 2, with positive recency in other cases.

An alternative prediction posits that the similarity of the effects observed in decisions under initial ignorance and numerical descriptions, in Pilot Study 2, stems from the participants' limited understanding of numerical descriptions. Presenting clear verbal descriptions may eliminate deviations from maximization.

3.3. Study 3: Reaction to Guidance and Misguidance

The iSampler model predicts similar sensitivity to guidance (nudges) and misguidance (misinformation). To evaluate this hypothesis, Study 3 will use the methodology of Study 2 (Figure 2's repeated decisions paradigm with verbal descriptions) to examine the choice task shown on the left-hand side of Figure 5.

Figure 5: iSampler predictions of the maximization-rate (A-rate) as a function of time (100 trials) with guidance, misguidance, and no description (control).

<table>
<tr><td></td><td colspan="2">**Predicted maximization rate (A-rate) as a function of time**</td></tr>
<tr><td>Payoff distributions</td><td>Full Feedback</td><td>Partial Feedback</td></tr>
<tr><td>A.
10% to win 50,
90% to lose 2 (EV = +3.2)
B.
50% to win 1,
50% to lose 1 (EV = 0)</td><td>
</td><td>
</td></tr>
</table>

Design, Procedure, and Participants: The experiment will employ a 3 (description conditions) x 2 (feedback conditions) between-group design. The three description conditions are: (1) No Description (Control): Participants receive no additional guidance or information. (2) Guidance: The EV-maximizing option (A) is described as "Better on average." (3) Misguidance: The low-EV option (B) is described as "Better most of the time."

These description conditions will be tested under two feedback conditions: (1) Full Feedback: Participants receive information about both the obtained payoff and the foregone payoff of the unchosen option, replicating the feedback method in Pilot Study 2. (2) Partial Feedback: Feedback is restricted to the obtained payoff only, with no information about foregone outcomes. Two-hundred participants will be assigned to each of the 6 groups.

Predictions: As demonstrated in Figure 5, iSampler predicts near indifference in the last 50 trials of the control group under full feedback. Guidance increases the maximization rate by 8%, while misguidance decreases it by 8%. Partial feedback reduces the maximization rate in the control condition and exacerbates the negative impact of misguidance.

Notice that iSampler's predictions are "pessimistic," as they imply that people will not learn to trust guidance more than they trust misguidance. Under an alternative hypothesis—used by Elon Musk to justify removing much of the accuracy checking in social media platform X that he owns—experience will increase the impact of guidance and eliminate the impact of misguidance. The predicted difference between the full and the partial feedback conditions suggests that iSampler captures the hot stove effect (Denrell & March, 2001). This effect implies that when the feedback is limited to the obtained payoffs, experience increases risk aversion.

3.4. Study 4: Comparison of alternative models of the joint impact of description and experience

As noted above, it is possible that the winners in the choice prediction competitions proposed in Study 1 will be pure large language models that do not model the underlying processes. In contrast, I believe that modeling the underlying processes will be crucial in predicting the joint impact of description and experience. My hypothesis rests on the assumption that people tend to select the strategy that led to the best outcomes in similar situations in the past. Thus, when the decision makers' past experiences are not known (to the model developer), as in the case of the one-shot decisions from description tasks considered in Study 1, the effort to model the cognitive processes adds more noise than value. Pure large language models minimize this noise by approximating these past experiences based on large data sets. I predict that when past experiences can be inferred, as in decisions from experience, cognitive models like iSampler that capture the way people process information obtained in these experiences add more value than noise.

To test this prediction, I plan to organize a second open-choice prediction competition using the 5-step format described in Table 2. The **focus** will be on the repeated choice paradigm described in Figure 2 (with full or partial feedback), and choice tasks drawn from the space examined in Study 1. The prediction goal will be the raw choice rates in four blocks of 25 trials, and the four contingent rates described in Figure 3 for each task.

Training and target experiments (Participants, Design, Task, and Procedure). The training and target experiments will use the method of Study 3. Each experiment (training and target) will examine 1200 tasks, with 400 tasks for each of three types of descriptions (Numeric, Verbal by ChatGPT, and Verbal by humans). Each participant in the main study will face one task for 100 trials, and each task will be faced by 20 participants. Thus, the total number of participants will be (2)(3)(400)(20) = 48,000 in the main experiments, and (2)400/5 = 160 in the preliminary verbal description experiments.

Baseline model and predictions. The baseline model will be iSampler. I predict that the winner will be a hybrid model that refines iSampler using machine learning tools. However, there are two interesting alternative hypotheses: The first states that the winner will be a model that generalizes prospect theory or traditional reinforcement learning models (e.g., Sutton & Barto, 1998), by using machine learning and large language models. A second alternative hypothesis states that the winner will be a pure large language model (as asserted by Binz et al., 2024).

The call will be similar to the call proposed in Study 1: It will be posted six months before the submission deadline in leading email lists in decision research, behavioral economics, machine learning, and reinforcement learning. It will include a link to the competition website. The site will feature a paper summarizing the competition focus, method, results of the training study, and the baseline model. Additionally, the site will include the raw data and participation rules (deadline, submission method, and feasible format of the submitted models). The prize for the winner will include coauthoring the paper summarizing the competition.

Reference

Binz, M., Akata, E., Bethge, M., Brändle, F., Callaway, F., Coda-Forno, J., ... & Schulz, E. (2024). Centaur: a foundation model of human cognition. *arXiv preprint arXiv:2410.20268*.

Barron, G., & Erev, I. (2003). Small feedback-based decisions and their limited correspondence to description-based decisions. Journal of Behavioral Decision Making, 16(3), 215–233.

Denrell, J., & March, J. G. (2001). Adaptation as information restriction: The hot stove effect. *Organization science*, *12*(5), 523-538.

Diederich, A., & Busemeyer, J. R. (1999). Conflict and the stochastic-dominance principle of decision making. *Psychological Science*, *10*(4), 353-359.

Erev, I., & Barron, G. (2005). On adaptation, maximization, and reinforcement learning among cognitive strategies. Psychological Review, 112(4), 912–931. https://doi.org/10.1037/0033-295X.112.4.912

Erev, I., Ert, E., Plonsky, O., Cohen, D., & Cohen, O. (2017). From anomalies to forecasts: Toward a descriptive model of decisions under risk, under ambiguity, and from experience. Psychological Review, 124(4), 369–409. https://doi.org/10.1037/rev0000062

Erev, I., Ert, E., Plonsky, O., & Roth, Y. (2023). Contradictory deviations from maximization: Environment-specific biases, or reflections of basic properties of human learning?. *Psychological Review*, *130*(3), 640.

Erev, I., Roth, Y., & Sonsino, D. (2022). Decisions from valuations of unknown payoff distributions. *Decision*, *9*(2), 172.

Erev, I., Ert, E., Roth, A. E., Haruvy, E., Herzog, S. M., Hau, R., Hertwig, R., Stewart, T., West, R., & Lebiere, C. (2010b). A choice prediction competition: Choices from experience and from description. Journal of Behavioral Decision Making, 23(1), 15–47. https://doi.org/10.1002/bdm.683

Friedman, J. H. (2001). Greedy function approximation: a gradient boosting machine. *Annals of statistics*, 1189-1232.

Gandhi, N., Zou, W., Meyer, C., Bhatia, S., & Walasek, L. (2022). Computational methods for predicting and understanding food judgment. *Psychological Science*, *33*(4), 579-594.

Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2), 263–292.

Liu, Y. (2019). Roberta: A robustly optimized bert pretraining approach. arXiv preprint arXiv:1907.11692, 364.

Nevo, I., & Erev, I. (2012). On surprise, change, and the effect of recent outcomes. Frontiers in Psychology, 3(FEB), 24. https://doi.org/10.3389/fpsyg.2012.00024

Peterson, J. C., Bourgin, D. D., Agrawal, M., Reichman, D., & Griffiths, T. L. (2021). Using large-scale experiments and machine learning to discover theories of human decision-making. *Science*, *372*(6547), 1209-1214.

Plonsky, O., Apel, R., Ert, E., Tennenholtz, M., Bourgin, D., Peterson, J. C., Reichman, D., Griffiths, T. L., Russell, S. J., Carter, E. C., Cavanagh, J. F., & Erev, I. (2019). Predicting human decisions with behavioral theories and machine learning. ArXiv Preprint ArXiv:1904.06866.

Plonsky, O., Teodorescu, K., & Erev, I. (2015). Reliance on small samples, the wavy recency effect, and similarity-based learning. *Psychological Review*, 122(4), 621–647.

Sutton, R. S., & Barto, A. G. (1998). Introduction to reinforcement learning. MIT press Cambridge.

Thorndike, E. (1898). Some experiments on animal intelligence. *Science*, *7*(181), 818-824.

Wakker, P. P. (2010). *Prospect theory: For risk and ambiguity*. Cambridge university press.

Readme:

a1, pa1,a2: the pay of distribution of option A (win a1 with probability pa1; a2 otherwise

b1, pb1,b2: the pay of distribution of option B (win b1 with probability pb1; b2 otherwise

corrAB: The correlation between the draws that determine the payff from A and B

A_label, B_label: The descriptions of A and B (empty cells imply no description).

Oneshot: The predicted A-rate based on study of one shot decisions.

Forgone: 1 if the participant receive full feedback, 0 if the feedback was limited to the selected option

arate1: Arate in trials 1 to 25

arate2: Arate in trials 25 to 50

arate3: Arate in trials 51 to 75

arate4: Arate in trials 76 to 100

afoka: Arate following an OK A choice. OK means that the last observed payoff from B was not larger.

Afrega: Arate following a regretful A choice. Regretful means that the last observed payoff from B was larger.

Afregb: Arate following a regretful B choice. Regretful means that the last observed payoff from A was larger.

Afokb: Arate following an OK B choice. OK means that the last observed payoff from A was not larger.